

Institute of Actuaries of India

Subject SA1– Health and Care

March 2022 Examination

INDICATIVE SOLUTION

Introduction

The indicative solution has been written by the Examiners with the aim of helping candidates. The solutions given are only indicative. It is realized that there could be other points as valid answers and examiner have given credit for any alternative approach or interpretation which they consider to be reasonable

Solution 1:

i) The heads are:

Opening Embedded Value, plus
 Expected return on free surplus, plus
 Return on in-force business,
 (-----Expected return
 -----Experience variances
 -----Operating assumption changes)
 Plus
 New business contribution, Less
 Development costs
 = Operating return before tax and exceptional items
 Plus
 Investment return variances, Plus
 Effect of economic assumption changes, Plus
 Exceptional items
 = Return on EV before tax
 Less, Attributed tax
 = Return on EV after tax
 Plus Capital raised
 Less Capital distributed
 = Closing Embedded Value

[Max 5]

ii) Expected return on free surplus

The risk-free investment return assumption at the start of the year is applied to the free surplus at the start of the year. [1]

The free surplus is assets less the Statutory reserve less the SCR. [½]

Return on in-force business--- expected return

This is the risk-free rate applied to the opening VIF. [½]

It is sometimes called the unwind of the discount rate'. [½]

It represents the value of the expected future profits being closer in time at the end of the analysis period, and thus being discounted by less and so having a higher value. [1]

Return on in-force business--- experience variances

The basic approach to determining the effect of experience variances on the net assets would be to: Roll forward the assets backing reserve to the end of the year using the beginning of the year BE assumptions as the expected experience over the year. [½]

As this roll forward is performed using best estimates of the experience, the assets and the BEL should still be equal at the end of the year. [½]

Repeat the roll forward process but change one of the items of experience from the expected value to its actual value. [1]

The change in the surplus at the end of the year gives the contribution from the item of experience. [½]

Repeat this process changing each item of experience in turn to identify its contribution to the change in net assets. [½]

Under this approach the experience variance from each source depends on the order in which each source is analysed. [½]

The process for performing this analysis is the same as that carried out for the analysis of surplus, ie starting from the opening position and changing one item at a time. [½]

The company will also need to consider the impact of experience variances on the other components of the EV. [½]

If actual experience is different from expected, then the VIF, RM and SCR components of the EV will also change. [1]

So, an experience variance item will also be required for any difference in the actual releases of profit and capital to those expected. [½]

Return on in-force business--- operating assumption changes

These will merge if the BE assumptions have changed [½]

There will be separate operating assumption changes to consider for the EV experience (projection) basis. [1]

Assumptions underlying the risk margin or capital requirement calculations might also change (eg correlation coefficients), which could also impact the EV. [1]

The assumed 'cost of capital' factor might also change (ie the parameters used to determine the cost of holding the RM and SCR), the impact of which would also need to be analysed. [½]

New business contribution

This is calculated as an increase in VIF plus an increase in net assets at the point of sale (assuming the new business is written on terms that are expected to be profitable). [1]

This could be assessed by simply taking the difference between the EV with and without the inclusion of model points that relate to new business written over the year. [1]

Investment return variances

In respect of the own funds, this item will include the difference in the change in assets backing the own funds over and above the assumed risk-free returns. [1]

Another component will be the impact of the assets moving differently to the movement in BEL. [½]

The actual return must be analysed against the risk-free rate plus any matching adjustment or volatility adjustment. [1]

There may also be an impact on the cost of holding the risk margin and additional capital. [½]

For example, the capital requirements may change due to actual investment returns being different from expected. [½]

In addition, there could be a contribution to the investment return variance if the risk-free yield curve moves differently to the asset yields. [½]

Investment return variance on the VIF will reflect the actual investment return differing from the discount rate being used to discount the future profits being taken credit for in this component). [½]

Effect of economic assumption changes

This will be determined in a very similar way to the investment return variance but will consider the impact of changing the assumptions for future economic experience (whereas the investment return variance considered the impact of having different economic experience than expected last year). [½]

It may be that some of these changes are combined with investment return variances, due to the close relationship between actual economic outcome over the year and the best estimate future economic assumptions, particularly in relation to yields and bond value movements. [1]

[Max 10]

iii) There will only be any impact if the company's expectation that the focus on IP sales will significantly boost sales is borne out in reality. [1]

There will be no impact on the current EV as that does not take account of new business. [1]

Assuming that the business is normally written to be profitable on a best estimate basis, as would be expected to be the case, the future EV would be higher than otherwise. [1]

This will increase the new business contribution in the analysis of change in EV ... [½]

...although the effect of the marketing campaign specifically would not be separately identified. [½]

An increase in sales will not directly affect in-force business experience. [½]

However, if future new business is expected to increase then the company may reduce the per-policy expense assumptions. [1]

This would increase the EV ... [½]

... and would come through in the analysis as an operating assumption change. [½]

It is unlikely that there will be a material impact on the experience variances, ... [½]

...although a small expense surplus may occur due to higher policy numbers to cover fixed expenses. [½]

It is possible that the future mix of business will change, ... [½]

... for example, the average policy size and the demographic mix may change, ... [½]

...but it will take some while before the full impact on the target market will be known. [½]

The cost of the marketing campaign will be included in the EV. [½]

The marketing campaign will cost money, pushing down assets which will show up in the EV. [½]

This may be treated as an exceptional item or a development cost in the analysis. [½]

The company may have to spend on systems and staff to deal with the increased business volumes. [½]

This spending would also reduce the EV and would partially offset the expense surplus items described above. [½]

Since the selling of impaired life annuity is stopped already, it will not impact the current & projected future EV [1]

[Max 10]

- iv) The first thing to consider is the assets allocated in respect of the business concerned. [1]
- If the two calculations were based on different assets, the values would be different for this reason. [½]
- For example, if the EV were based on 100 million assets and 105 million were transferred to the new insurer, this fact alone would lead you to expect a payment in respect of the business bigger than the EV by 5 million.) [½]
- Consider how the sale price will be negotiated. It is likely that the purchaser and seller will both agree to use EV calculations to determine how much should be paid for the business ... [½]
- but if the sale price is agreed by some other means, it is not possible to comment on its likely size in comparison with the EV. [½]
- If the EV is to be used, then there will be a need to consider the basis for the calculations. [½]
- The sale basis is a question of negotiation. The purchaser (seller) will favour a basis that produces a lower (higher) EV. [1]
- Whether this results in a higher or lower EV than one using the published EV basis will depend on the strength of the negotiating parties. [½]
- Some of the features that may affect this are as follows:
- The sale of the business is not a 'forced' sale. The seller is therefore not in a weak position and may be able to achieve a reasonable price for the business. Following this logic, the sale price may exceed the EV. [1]
- Conversely, if the withdrawal from the market has a specific driver, eg changing capital requirements of this class of business under Solvency II, the seller may be in a weaker position and not achieve as good a price. [½]
- The published EV is market-consistent so should in principle give values achievable when selling a block of business. [1]
- The purchaser may base the negotiations on their own expense levels rather than the Seller's [1]
- So, if the sale is achieved to a very low expense company, a price higher than the EV may be achieved because of the lower expenses expected to be incurred by the purchasing company. [½]
- It is likely that any such synergy would be shared. [½]
- On the other hand, the sale price will need to allow for transaction costs ... [½]
- ... and the costs of the new company taking the business on board. [½]

It may be argued in negotiations that the event of selling the business will cause a surge of lapses / surrenders. [1]

The purchasing company may insist on a very cautious (ie big) allowance for lapses in the sale price. [½]

This might make the sale price lower than the published EV. [½]

There may be an argument from the purchaser that there will be greater uncertainty regarding the experience of the business post-sale. [1]

The purchaser may therefore push for greater allowances for risk margins and required capital, resulting in a lower EV. [1]

Another issue is that the published EV will be based on assumptions set at a single point in time each year. The sale price is a one-off that may be determined between annual reporting dates. [½]

The sale price may therefore be higher or lower because of unusual current conditions (eg unusually high or low interest rates). [1]

Overall, with factors going in different directions, it is not possible to say which of the figures will be larger. [½]

[Max 10]

- v) The EV movement analysis shows the net impact of deviation in current surplus and future profitability and the surplus analysis shows the impact of deviation on current profits only. [1]

A single head might have opposite impact on EV & Surplus. For example, increase in lapse rates might have positive impact on surplus but negative impact on EV. [2]

The surplus analysis will show the financial effect of divergences between actual experience and that assumed in the valuation basis. [½]

This will help with a review of the valuation assumptions, exposing which assumptions are the most financially significant. [½]

It must be remembered though, that valuation assumptions are long-term and should not be changed hastily in the light of just one year's results. [½]

Generally, it will show the financial effect of writing new business. [½]

If the analysis is carried out independently of the valuation data, it can provide a check on both the valuation data and the valuation process. [½]

It will help to identify any causes for concern, so that corrective action may be taken. [½]

The trends in the items of surplus may give useful information on trends in the experience of the company. [½]

[Max 5]

- vi) For long-term insurances, the analysis of surplus looks at the surplus arising each year throughout their lifetime. [½]

For existing business at least, there may be little chance to react to adverse experience since the premium rates are usually fixed. [½]

For PMI, there is the opportunity at renewal to change prices, product design, underwriting or claim conditions ... [½]

... and hence it is useful to analyse profit more regularly throughout the year. [½]

An analysis of surplus would normally be an annual exercise for a long-term insurer. [½]

For a PMI insurer, the company would again assess the profitability of business by yearly cohort, but would also perform quarterly analyses to establish up-to-date positions. [½]

The quarterly calculations will be fed through to the price being charged for new business, to take account of the most recent changes in claim trends. [½]

The key items to monitor are:

rates of incidence of claim, ie the number of separate claims divided by the exposed to risk [½]

average cost per claim, ie the total incurred cost of claims divided by the number of separate claims

loss ratio, ie the total incurred cost of claims divided by the relevant earned premium. [½]
 In addition to these, analysis of persistency is also important, to assess the profits lost through withdrawals. [½]
 Analysis of investment profit is not so important for the short-term business. [½]
 Whilst the loss ratio is quick and simple to calculate, any trends in either frequency or size of claim will be hidden. [½]
 Therefore, if data allows, it is more useful to monitor the first two indices. [½]
 Each will be calculated for the cohorts of business under review and will be compared with the assumptions made in the original premium pricing. [½]
 Cohorts by duration are often reviewed since many PMI portfolios exhibit a selection effect; ... [½]
 ... the loss ratio typically increases by duration, depending on the method and depth of medical underwriting at outset. [½]
 Incurred claims will include allowance for the total ultimate expected claim cost and so must include all claims paid to date (whether settled or not), and all claims yet to be reported. [1]
 The definition of a claim should be consistent so that trends in experience can be meaningfully analysed, and results compared to those of competitors. [1]
 If no claims discount systems are in operation, it is crucial to be able to assess experience by cohort accurately and quickly, in order to fine-tune the methodology of bonus-malus and promotion / demotion scales in operation. [1]

[Max 10]

[50 Marks]

Solution 2:

i)

- Solvency ratio refers to the ratio of available capital to required capital
- Regulators require insurers to always maintain certain level of solvency ratio
- Solvency ratio of <100% => insolvency
- Typically, there is a threshold solvency ratio, say 120% - when solvency ratio tips below that level, the regulator starts to intervene
- Both available capital and required capital are determined based on the regulatory capital regime applicable
- Required capital tends to reflect the risk profile – higher the risk greater the capital required
- Risk profile includes biometric risks (mortality/morbidity/longevity) as well as others including
- Investment – both in absolute terms and in relation to the extent of asset/liability mismatch
- Operational – fraud, data loss, people, system, and other operational failures etc.
- Counterparty – failure of third parties
- There is also typically minimum capital requirement for doing insurance business
- So, the required capital is then Max (minimum capital, required capital)
- Available capital is the excess of assets over the liabilities
- Assets here refer to the value of assets to the extent admissible for regulatory solvency purpose
- Liabilities are the value of policy obligations calculated as per regulatory guidance/rules

[½ per point; Max 5]

ii)

- It is likely that the regulatory intervention will kick in – that might include
 - Limiting new business/closure to new business
 - Immediate requirement to pump in capital by the shareholders

- Bad publicity => lower sales
- Brokers not promoting the company products => lower sales
- Lower sales => poor contribution to company overheads => worsening the financials
- If the company is traded, drop in share price
- Any company expansion plans will likely be stalled
- Potential downgrading of Credit rating
- Potential impact on salary increase/bonus payout to staff
- Likely increase in staff turnover
- Likely increase in salesforce turnover
- Likely impact on staff morale => affecting customer service quality
- Increased lapse/surrender => again, worsening the financials

[½ per point; Max 5]

iii)

- Look at the drivers of solvency (available capital and required capital) from both asset and liability sides
- Any asset admissibility issues due to domiciling of assets, type of assets etc.?
- Can they be addressed by reallocation of assets?
- Need to consider cost implications of sale/purchase that might be involved
- Reserving assumptions more prudent than required by regulations?
- Claims outgo larger than expected (large claims/catastrophe/pandemic etc.) – one-off or recurring
- Could the results have been better with a different reinsurance structure?
- Cost/benefit implications of such a reinsurance?
- Impact of different reinsurance structures on the required capital
- While more reinsurance might help reduce capital requirement...
- ..it might increase the counterparty risk and hence increased capital
- Profit testing of the products – are we charging enough?
- Expense analysis – are there areas to get more efficient?
- ..would outsourcing of some of the activities be more efficient?
- Shareholder's appetite to invest more
- .. especially if the solvency challenge is driven by large and sudden growth in volumes
- ...or, caused by change in accounting rules/practices and it is a transition issue
- ...or, new capital-intensive product sold a lot
- Customer persistency analysis – is higher or lower than expected withdrawals causing any strain?
- Analysis of new business strain...
- ...would change of sales incentive structures help?
- Impact of financial reinsurance arrangement (if permitted by the regulations)
- Analysis of ALM position and solvency implications
- Any operational issues – fraud, waste etc. and how they can be fixed

[½ per point; Max 9]

iv)

- Summary of analyses done outlining
- ...the key causes of solvency drop
- ...systemic issue or temporary
- Business plan for the next few years and its impact on solvency
- List of actions that will be taken with rationale for each

- ...and the impact of them on the solvency
- ...timelines for execution of each action item
- ...projected solvency position over the next few years as per regulatory solvency rules
- ...certified by the Appointed Actuary
- ...regular monitoring and internal governance/regulatory reporting
- Implications of the borderline solvency on PRE and how they would be addressed
- ... how will the customer/investor confidence be maintained/restored
- Would the plan be made public and why?
- Why no regulatory action necessary?
- Board approval for the plan

[½ per point; Max 6]

v)

- Expense is an item of outgo for the insurer
- ...and, expense overrun could be eating into available capital
- ...this is typically a challenge for small/medium size insurers
- ...especially in early years when there is no economies of scale
- Risk based capital framework includes expense risk (including expense inflation)
- ..so, will be a driver of required capital
- Even otherwise, expense management is a key element for prudent management of an insurer and driver of profitability and hence capital

[½ per point; Max 3]

vi)

- Acquisition expenses
 - relate to sales and marketing, underwriting (including expenses related to persistency management)
 - unit measurements include no. of applications, no. of policies issued, face amount
- Administration expenses
 - relate to the ongoing policy-servicing costs (including premium collections, operational management, monitoring and reporting)
 - unit measurements include no. of policies inforce, premium income, no. transactions
- Claim expenses
 - relate to validating and paying claims (including any legal costs associated for disputed claims)
 - unit measurements include no. of claims, size of claim, no. of cancellation/refund processed
- Investment expenses
 - relate to managing the insurer's investment portfolio
 - unit measurements include market value, acquired value, sale value
- Overhead expenses
 - Other costs not included elsewhere, including executive management, actuarial, legal, HR, IT, etc.
 - Unit measurements include required capital, no. of employees, premium volume, percentage of non-overhead expenses

[1½ per expense type; Max 6]

vii)

a) Individual Income Protection

- Long-term product - initial expenses involve intermediary commission, underwriting and setting up policy in the admin systems
- Commission and admin – mostly directly linked to policies issued (no & size) – overheads linked to volume and so check if sales are in line with expected
- Cost of underwriting driven by applications, and it will need to be spread over the policies
- so, track the placement ratio (application to policy)... lower than placement than expected higher the expenses than expected...
- multiple claims on the same policy – so, incidence as well as termination rates drive the expenses associated...
- ..so, monitor if actual claims (incidence as well as termination) are in line with expected claims – deviation could point to expense gap
- Expense inflation is a key item for this product... check if the actual inflation is in line with that allowed for in the pricing

[½ per point; Max 3]

b) Group Medical Reimbursement

- Short-term product – group and so less underwriting – so, initial expense is largely commission and policy setup
- ...vary between new business and renewals – so, sales and persistency will drive the spread of overheads
- Significant claim expenses – linked to number and size – will depend upon inhouse vs outsourced to a TPA
- inhouse => higher volume/complexity of claims and lower business volume would invalidate the expense assumption => so, check actual to expected
- Similarly, TPA fee linked to number of members/volumes will depart from pricing if the numbers differ
- expense allowance for other overheads will again be linked to volume
- expense inflation is less of an issue with this being short-term

[½ per point; Max 3]

c) Group Critical Illness

- Short-term product – so, initial expense is largely commission and policy setup – key is check if volume sufficient to cover fixed overheads
- Some underwriting will be involved if this is voluntary...
- so, level of participation is important to monitor.. lower than expected participation => lower contribution to expenses than expected
- Less likely to be stand-alone and typically attached to group life cover – so, the rider take-up will drive this benefits contribution to the full product
- ..check actual to expected take-up... the lower the take up the lower the contribution to expenses
- Claim expense linked to number (and to some extend size) of claims... so, monitor claim experience
- ...higher than expected incidence of claims => higher than expected expenses

[½ per point; Max 3]

viii)

- Suitability will depend upon the nature of the problem that the insurer is facing – what drives the low solvency

- Also, whether or not the solution is acceptable to the regulator
- ..and the cost vs benefit of the solution
- There are different FinRe solutions/structures...
- Lump sum fund in return for profit from reinsurance
- ...Assets created by advancing of funds to the insurer by the reinsurer
- ...with the repayment contingent upon surplus from reinsurance in future years
- ...so, no liability created thereby improving assets over liabilities
- Value of Inforce monetization
- ...recognizing profit embedded in the inforce blocks through reinsurance
- ...works when you have large blocks of profitable long-term insurance business
- ...which is less likely in this situation – medium size health insurer
- Commission financing
- ...helps reduce new business strain
- ..again, works for long-term business with significant initial commission/underwriting and other acquisition costs
- Assets restructuring
- ...through which attempts to reduce the inadmissibility of assets, for example of due to asset domiciling requirements of the regulator
- Overall, given the nature of this case, it is less likely that FinRe solutions might help here

[½ per point; Max 7]

[50 Marks]
